

## 9.7 Polynomial Matrices and the Smith Normal Form

The rational canonical form of the previous section is a complete answer to the similarity problem, but the proof did not tell us how to *find* the invariant factors of a given matrix. This section supplies the missing algorithm, and it comes from an unexpected place: instead of working with  $A$ , we work with the single polynomial matrix  $xI - A$ , and we reduce it by row and column operations whose scalars are allowed to be polynomials. The reduced form is diagonal, its non-constant diagonal entries are the invariant factors, and two matrices are similar exactly when their reduced forms agree. On the way we finally pay off a promise from Chapter 3: every square matrix is similar to its transpose, over every field.

### 9.7.1 The space as a module over the polynomials

Everything in this chapter is built by applying **polynomials** in  $T$  to vectors: the  $T$ -annihilator of a vector, the cyclic subspace  $Z(\mathbf{v}; T)$ , the kernels  $\ker p(T)$ . Name that structure once. For  $V$  a vector space over  $F$  and  $T \in \mathcal{L}(V)$ , the product

$$F[x] \times V \rightarrow V, \quad p \cdot \mathbf{v} := p(T) \mathbf{v}$$

satisfies, by Theorem 5.7.1, exactly the vector space axioms with  $F$  replaced by  $F[x]$ : it is additive in each factor,  $(pq) \cdot \mathbf{v} = p \cdot (q \cdot \mathbf{v})$ , and  $1 \cdot \mathbf{v} = \mathbf{v}$ . A set with an addition and a product by the elements of a commutative ring  $R$  obeying these rules is an  **$R$ -module**, so the pair  $(V, T)$  is the same thing as an  $F[x]$ -module that is finite-dimensional over the constants  $F$ . We develop no general theory of modules; the point is the dictionary:

| $F[x]$ -module language                                               | operator language                                     |
|-----------------------------------------------------------------------|-------------------------------------------------------|
| submodule of $V$                                                      | $T$ -invariant subspace (Definition 8.1.1)            |
| submodule generated by $\mathbf{v}$ , that is $F[x] \cdot \mathbf{v}$ | cyclic subspace $Z(\mathbf{v}; T)$ (Definition 9.5.1) |
| the annihilator of $\mathbf{v}$ in $F[x]$                             | the $T$ -annihilator of $\mathbf{v}$                  |
| $V$ is generated by one element                                       | $T$ has a cyclic vector                               |

A subspace closed under multiplication by  $x$  is closed under multiplication by every polynomial, which is why the first two lines match.

Where does  $xI - A$  come from here? Fix  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  with  $A = [T]_{\mathcal{B}}$ . As an  $F[x]$ -module,  $V$  is generated by  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , subject to the  $n$  relations  $x \cdot \mathbf{b}_j = \sum_i a_{ij} \mathbf{b}_i$ , whose coefficient vectors are the columns of  $xI - A$ . So  $xI - A$  is the **matrix of relations**, and changing generators or replacing one relation by a combination of relations should not change  $V$ . That is the intuition behind the operations we now allow; the proofs below are matrix proofs and do not rely on it.

### 9.7.2 Row and column operations with polynomial multipliers

A matrix in  $M_n(F[x])$  will be called a **polynomial matrix**. Over a field we reduced a matrix by three kinds of row operation (Definition 2.2.1). Over  $F[x]$  we must be careful with one of

them: multiplying a row by  $x$  cannot be undone, because  $x$  has no inverse in  $F[x]$ . So the scaling operation keeps its scalar in  $F$ , while the operation that adds a multiple of one row to another may use any polynomial multiplier.

### Definition 9.7.1 — Elementary Operations over $F[x]$

Let  $M \in M_n(F[x])$ . An **elementary row operation** over  $F[x]$  on  $M$  is one of:

- (E1) interchanging two rows;
- (E2) multiplying one row by a **non-zero scalar**  $c \in F$ ;
- (E3) adding  $h$  times one row to a **different** row, where  $h \in F[x]$  is **any** polynomial.

**Elementary column operations** over  $F[x]$  are defined the same way with “column” in place of “row”.

In words: (E1) and (E3) are as over a field, except that (E3) now permits a polynomial multiplier; (E2) is restricted, because only the non-zero constants have inverses in  $F[x]$ . Each of the three is reversible by an operation of the same kind: swap back, scale by  $c^{-1}$ , add  $-h$  times the row back.

Two polynomial matrices reachable from one another by such operations should count as the same. As over a field (Definition 3.8.1), the clean way to say this is with invertible matrices on both sides — but “invertible” must now mean invertible *inside*  $M_n(F[x])$ .

### Proposition 9.7.1 — Invertible Polynomial Matrices

Let  $U \in M_n(F[x])$ . Then  $U$  has an inverse in  $M_n(F[x])$  if and only if  $\det U$  is a **non-zero constant**.

*Proof.* ( $\Rightarrow$ ) Suppose  $UW = I$  with  $W \in M_n(F[x])$ . Taking determinants and using multiplicativity over the commutative ring  $F[x]$  (Theorem 6.5.4 and the remark on commutative rings after it),  $\det U \cdot \det W = 1$ . By Theorem 0.8.3,  $\deg(\det U) + \deg(\det W) = \deg 1 = 0$ , and degrees of non-zero polynomials are  $\geq 0$ , so  $\deg(\det U) = 0$ ; that is,  $\det U \in F \setminus \{0\}$ .

( $\Leftarrow$ ) Suppose  $\det U = c \in F \setminus \{0\}$ . The adjugate identity

$$U \operatorname{adj} U = \operatorname{adj} U U = (\det U) I$$

holds over any commutative ring (Theorem 6.6.2 and the remark on commutative rings following it), and  $\operatorname{adj} U \in M_n(F[x])$ , since its entries are minors of  $U$ . So  $W = c^{-1} \operatorname{adj} U \in M_n(F[x])$  satisfies  $UW = WU = I$ .

■

### Definition 9.7.2 — Equivalence over $F[x]$

Let  $M, M' \in M_n(F[x])$ . We say  $M$  and  $M'$  are **equivalent over**  $F[x]$ , written  $M \approx M'$ ,

if there are  $U, V \in M_n(F[x])$ , **each invertible in**  $M_n(F[x])$ , with

$$M' = UMV.$$

### Proposition 9.7.2 — Operations and Equivalence

Let  $M, M' \in M_n(F[x])$ .

- (a)  $\approx$  is an equivalence relation on  $M_n(F[x])$ .
- (b) If  $M'$  is obtained from  $M$  by a finite sequence of elementary row and column operations over  $F[x]$ , then  $M \approx M'$ .

*Proof.*

- (a) Take  $U = V = I$  for reflexivity. If  $M' = UMV$ , then  $M = U^{-1}M'V^{-1}$ , and  $U^{-1}, V^{-1}$  are invertible in  $M_n(F[x])$ ; this gives symmetry. If  $M' = UMV$  and  $M'' = U'M'V'$ , then  $M'' = (U'U)M(VV')$ , and a product of invertible matrices is invertible, which gives transitivity.
- (b) Each elementary row operation over  $F[x]$  is left multiplication by the matrix obtained by performing that operation on  $I_n$ , exactly as over a field (Theorem 2.4.1, whose proof uses only the ring operations). Those three matrices are  $P_{ij}$ ,  $D_i(c)$  and  $I + hE_{ij}$  with  $i \neq j$ , of determinants  $-1$ ,  $c \neq 0$  and  $1$  respectively (Theorem 6.4.6 and Theorem 6.5.1, valid over a commutative ring), hence invertible in  $M_n(F[x])$  by Proposition 9.7.1. Column operations are right multiplication by the same matrices. Now use (a).

■

**Warning.** Multiplying a row by a non-constant polynomial is not an allowed operation. It changes the answer. The matrix  $M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  becomes  $\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}$  after “multiplying row 1 by  $x$ ”, and the two are not equivalent: their determinants are 1 and  $x$ , and by Proposition 9.7.1 the first is invertible over  $F[x]$  while the second is not. The rule to remember is that only the **units** of  $F[x]$  — the non-zero constants — may be used as scaling factors.

## 9.7.3 The Smith normal form

*Every polynomial matrix can be brought to a diagonal form whose diagonal entries divide one another in turn, and that form is unique.*

The existence proof is Euclidean division, run on whichever non-zero entry has the smallest degree: if that entry fails to divide one of its neighbors in the same row or column, the remainder gives a non-zero entry of strictly smaller degree, and degrees cannot decrease forever.

### Theorem 9.7.3 — Smith Normal Form

Let  $F$  be a field,  $n \geq 1$  and  $M \in M_n(F[x])$ . Then:

- (a) **(Existence)**  $M$  is equivalent over  $F[x]$ , by a finite sequence of elementary row and column operations, to a diagonal matrix

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

in which every  $d_i$  is **monic or zero** and  $d_i \mid d_{i+1}$  for  $i = 1, \dots, n-1$ .

- (b) **(Uniqueness)** The list  $(d_1, \dots, d_n)$  is determined by  $M$ : if  $M \approx \text{diag}(d_1, \dots, d_n) \approx \text{diag}(e_1, \dots, e_n)$  with both lists as in (a), then  $d_i = e_i$  for every  $i$ .

The matrix  $D$  is called the **Smith normal form** of  $M$ , and  $d_1, \dots, d_n$  its **invariant divisors**.

Since  $0 \mid f$  holds only for  $f = 0$ , the divisibility chain forces the zero entries, if any, to be the last ones.

*Idea. Step roadmap for existence.* ① Among all matrices reachable from  $M$  by operations, pick one whose smallest non-zero entry degree is least, and move that entry to the corner. ② That corner entry  $f$  must divide every other entry of its row and column: otherwise divide with remainder and produce an entry of smaller degree, contradicting the choice in ①. ③ Clear the first row and column with  $f$ ; the same remainder trick, after adding a row to the first one, shows  $f$  divides every entry of what is left. ④ Induct on the size. The divisibility  $d_1 \mid d_2$  survives the induction because operations replace entries by  $F[x]$ -combinations of entries.

For uniqueness, find a quantity that can be read off both  $M$  and  $D$ . The greatest common divisor of all  $k \times k$  minors is such a quantity: minors of a product are  $F[x]$ -combinations of minors of the factors (Cauchy–Binet), so the gcd cannot change under equivalence, and for a diagonal matrix with a divisibility chain it is simply  $d_1 d_2 \cdots d_k$ .

*Proof.*

- (a) If  $M = 0$ , it is already of the stated form. So assume  $M \neq 0$ .  
For a non-zero  $N \in M_n(F[x])$  put

$$\delta(N) := \min\{\deg n_{ij} : n_{ij} \neq 0\} \in \mathbb{N},$$

the least degree of a non-zero entry. Let  $\mathcal{S}$  be the set of all non-zero matrices obtainable from  $M$  by finite sequences of elementary row and column operations over  $F[x]$ ; it contains  $M$ . The set  $\{\delta(N) : N \in \mathcal{S}\}$  is a non-empty subset of  $\mathbb{N}$ , so it has a least element; **fix**  $N \in \mathcal{S}$  **with**  $\delta(N)$  **least**.

**Step 1. Put a least-degree entry in the corner, monic.** Choose  $(i, j)$  with  $n_{ij} \neq 0$  and  $\deg n_{ij} = \delta(N)$ . Swapping row  $i$  with row 1 and column  $j$  with column 1 (operations (E1)), and then scaling the new first row by the inverse of the leading coefficient of the corner entry (operation (E2)), we obtain a matrix in  $\mathcal{S}$  whose  $(1, 1)$ -entry  $f$  is **monic** of degree  $\delta(N)$ . These operations permute the entries and rescale some of them, so the new matrix still has least non-zero entry degree  $\delta(N)$ . Rename it  $N$ .

**Step 2.  $f$  divides every entry of the first row and the first column.** Suppose  $f \nmid n_{1j}$  for some  $j \geq 2$ . By Theorem 5.1.1,  $n_{1j} = qf + r$  with  $r \neq 0$  and  $\deg r < \deg f$ . Adding  $-q$  times column 1 to column  $j$  (operation (E3)) produces a matrix  $N' \in \mathcal{S}$  whose  $(1, j)$ -entry is  $r \neq 0$  with  $\deg r < \deg f = \delta(N)$ . Then

$\delta(N') < \delta(N)$ , contradicting the minimality of  $\delta(N)$ . Hence  $f \mid n_{1j}$  for every  $j$ , and the same argument with rows in place of columns gives  $f \mid n_{i1}$  for every  $i$ .

**Step 3. Clear the first row and column.** For  $j \geq 2$  add  $-(n_{1j}/f)$  times column 1 to column  $j$ ; the multipliers are polynomials by Step 2, and after these operations the first row is  $(f, 0, \dots, 0)$  while the first column is unchanged. Then for  $i \geq 2$  add  $-(n_{i1}/f)$  times row 1 to row  $i$ ; since the first row is now  $(f, 0, \dots, 0)$ , this clears the first column and changes nothing else. The result is

$$\begin{pmatrix} f & 0 \\ 0 & M' \end{pmatrix} \in \mathcal{S}, \quad M' \in M_{n-1}(F[x]),$$

and its least non-zero entry degree is still  $\delta(N)$ : it is  $\geq \delta(N)$  by minimality, and  $f$  is an entry of degree  $\delta(N)$ .

**Step 4.  $f$  divides every entry of  $M'$ .** Suppose  $f \nmid (M')_{kl}$  for some  $k, l$ . Add row  $k+1$  to row 1. The new first row is  $(f, (M')_{k1}, \dots, (M')_{k,n-1})$ , so the  $(1, l+1)$ -entry of the resulting matrix  $N'' \in \mathcal{S}$  is  $(M')_{kl}$ , which is not divisible by  $f$ , while the  $(1, 1)$ -entry is still  $f$ , monic of degree  $\delta(N)$ . By minimality  $\delta(N'') = \delta(N)$ , so Step 2 applies to  $N''$  and says that  $f$  divides every entry of the first row of  $N''$  — a contradiction.

**Step 5. Induct on  $n$ .** We prove (a) by induction on  $n$ . For  $n = 1$ , scale the single entry to be monic (or leave it 0). For  $n \geq 2$ , Steps 1–4 give a sequence of operations taking  $M$  to  $f \oplus M'$  with  $f$  monic and  $f$  dividing every entry of  $M'$ . By the induction hypothesis there are operations taking  $M'$  to  $D' = \text{diag}(d_2, \dots, d_n)$  with every  $d_i$  monic or zero and  $d_2 \mid \dots \mid d_n$ ; performing the corresponding operations on rows and columns 2,  $\dots$ ,  $n$  of  $f \oplus M'$  takes it to  $f \oplus D' = \text{diag}(f, d_2, \dots, d_n)$ . It remains to see that  $f \mid d_2$ . By Proposition 9.7.2,  $D' = U'M'V'$  for some  $U', V' \in M_{n-1}(F[x])$ , so every entry of  $D'$  is a sum of products  $u'_{ik}(M')_{kl}v'_{lj}$  (Definition 0.9.9) and is therefore divisible by  $f$ , since every  $(M')_{kl}$  is. In particular  $f \mid d_2$ . Setting  $d_1 = f$  finishes the induction.

- (b) Call a set  $I \subseteq [n]$  with  $|I| = k$  a  **$k$ -set**. For  $M \in M_n(F[x])$  and  $1 \leq k \leq n$  define the  $k$ -th **determinantal divisor**

$$\Delta_k(M) := \text{the monic generator of the ideal } \langle \det M_{I,J} : I, J \text{ } k\text{-sets} \rangle \subseteq F[x],$$

and  $\Delta_k(M) := 0$  if all the  $k \times k$  minors of  $M$  vanish. This is well defined by Proposition 5.2.1 and Theorem 5.2.2: the ideal generated by the finitely many  $k \times k$  minors is  $\langle \Delta_k(M) \rangle$  for exactly one monic  $\Delta_k(M)$  unless the ideal is  $\{0\}$ . Equivalently,  $\Delta_k(M)$  is the monic greatest common divisor of the  $k \times k$  minors: it divides each of them, and every common divisor of them divides it (Theorem 5.2.3, extended from two polynomials to finitely many by the same ideal argument).

**Claim 1.**  $M \approx M'$  implies  $\Delta_k(M) = \Delta_k(M')$  for every  $k$ .

*Proof.* Let  $M' = UMV$  with  $U, V$  invertible over  $F[x]$ , and let  $I, J$  be  $k$ -sets. The submatrix of a product on rows  $I$  and columns  $J$  is the product of the submatrix of the left factor on rows  $I$  and all columns with the submatrix of the right factor on all rows and columns  $J$  (Definition 0.9.9, read off entry by entry). Applying this twice and then Theorem 6.7.2 over the commutative ring  $F[x]$  — legitimate by the remark after that theorem — to the  $k \times n$  times  $n \times k$  products involved,

$$\det M'_{I,J} = \sum_S \det U_{I,S} \det (MV)_{S,J} = \sum_S \sum_R \det U_{I,S} \det M_{S,R} \det V_{R,J},$$

with  $S$  and  $R$  running over the  $k$ -sets. So every  $k \times k$  minor of  $M'$  is an  $F[x]$ -combination of  $k \times k$  minors of  $M$ , hence lies in  $\langle \det M_{S,R} \rangle$ . Therefore  $\Delta_k(M)$  divides every  $k \times k$  minor of  $M'$ , and so  $\Delta_k(M) \mid \Delta_k(M')$ . Exchanging the roles of  $M$  and  $M'$  using  $M = U^{-1}M'V^{-1}$  gives  $\Delta_k(M') \mid \Delta_k(M)$ . Two monic polynomials dividing each other are equal (Proposition 5.1.2), and if one is 0 so is the other. ■

**Claim 2.** If  $D = \text{diag}(d_1, \dots, d_n)$  with  $d_1 \mid d_2 \mid \dots \mid d_n$ , then  $\Delta_k(D) = d_1 d_2 \cdots d_k$ .

*Proof.* Let  $I, J$  be  $k$ -sets. If  $I \neq J$ , pick  $i \in I \setminus J$ ; row  $i$  of  $D$  has its only possibly non-zero entry in column  $i \notin J$ , so the submatrix  $D_{I,J}$  has a zero row and  $\det D_{I,J} = 0$ . If  $I = J = \{i_1 < \dots < i_k\}$ , then  $D_{I,I} = \text{diag}(d_{i_1}, \dots, d_{i_k})$  and  $\det D_{I,I} = d_{i_1} \cdots d_{i_k}$  (Theorem 6.4.6, valid over the commutative ring  $F[x]$  by the remark after it). Since  $i_j \geq j$ , the chain gives  $d_j \mid d_{i_j}$  for each  $j$ , hence

$$d_1 d_2 \cdots d_k \mid d_{i_1} d_{i_2} \cdots d_{i_k}.$$

So  $d_1 \cdots d_k$  divides every  $k \times k$  minor of  $D$ , and it is itself the minor for  $I = J = \{1, \dots, k\}$ . Hence the ideal generated by the  $k \times k$  minors is  $\langle d_1 \cdots d_k \rangle$ . If  $d_1 \cdots d_k \neq 0$  it is monic, being a product of monic polynomials, and equals  $\Delta_k(D)$ . If  $d_1 \cdots d_k = 0$ , say  $d_m = 0$  with  $m \leq k$ , then for every  $k$ -set  $I$  we have  $i_k \geq k \geq m$ , so  $d_m \mid d_{i_k}$  forces  $d_{i_k} = 0$  and the minor vanishes; thus  $\Delta_k(D) = 0 = d_1 \cdots d_k$ . ■

Now let  $M \approx \text{diag}(d_1, \dots, d_n)$  with the  $d_i$  as in (a). By Claims 1 and 2,

$$\Delta_k(M) = d_1 d_2 \cdots d_k \quad (k = 1, \dots, n),$$

a quantity depending only on  $M$ . Reading this backwards recovers the  $d_i$ : first  $d_1 = \Delta_1(M)$ ; and for  $k \geq 2$ , if  $\Delta_{k-1}(M) \neq 0$  then  $d_k = \Delta_k(M)/\Delta_{k-1}(M)$ , while if  $\Delta_{k-1}(M) = 0$  then  $d_1 \cdots d_{k-1} = 0$ , so some  $d_m = 0$  with  $m \leq k-1$ , and the chain forces  $d_k = 0$ . Hence any second list  $(e_1, \dots, e_n)$  as in (a) satisfies the same recipe and equals  $(d_1, \dots, d_n)$ . This proves the theorem. ■

The proof is an algorithm, and it is the one to use by hand: hunt for the entry of least degree,

divide, and repeat.

**Smith normal form of  $M \in M_n(F[x])$ .**

1. If  $M = 0$ , stop. Otherwise find a non-zero entry of least degree and move it to position  $(1, 1)$  by a row and a column swap; scale row 1 so that the entry is monic. Call it  $f$ .
2. For each other entry  $g$  in row 1 or column 1 with  $f \nmid g$ , replace  $g$  by its remainder on division by  $f$  (subtract a multiple of column 1, resp. row 1), move the remainder to the corner, and go back to step 1.
3. When  $f$  divides all of row 1 and column 1, clear them. If some entry  $g$  of the remaining block is not divisible by  $f$ , add its row to row 1 and go back to step 2.
4. Repeat on the remaining  $(n - 1) \times (n - 1)$  block.
5. Check the answer:  $\Delta_k$  must equal  $d_1 \cdots d_k$ . In particular  $d_1 \cdots d_n$  is  $\det M$  rescaled by a non-zero constant to be monic, and  $d_1$  is the monic gcd of the entries.

*Quick check.* Two polynomial matrices in  $M_2(F[x])$  have determinants  $x^2$  and  $x^2$ , and in both the greatest common divisor of the four entries is 1. Are they equivalent over  $F[x]$ ?

*Solution.* Yes. Here  $\Delta_1 = 1$  (the gcd of the  $1 \times 1$  minors, that is, of the entries) and  $\Delta_2$  is the determinant rescaled to be monic, so  $\Delta_2 = x^2$  for both. By [Theorem 9.7.3](#) both have Smith normal form  $\text{diag}(1, x^2)$ , and since  $\approx$  is an equivalence relation ([Proposition 9.7.2 \(a\)](#)) they are equivalent to each other. For  $2 \times 2$  matrices the pair  $(\Delta_1, \Delta_2)$  is therefore a complete invariant.

### Example. (A Smith Normal Form Computed by Hand)

Let

$$A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Find the Smith normal form of  $xI - A$  over  $\mathbb{Q}[x]$ , naming every operation, and check the answer with determinantal divisors.

*Solution.* We have

$$xI - A = \begin{pmatrix} x-2 & -1 & 2 \\ -1 & x-2 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

*Find the corner.* The entries of least degree are the three entries equal to  $-1$ . Swap columns 1 and 2, then multiply the first row by  $-1$ :

$$\begin{pmatrix} -1 & x-2 & 2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -(x-2) & -2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

The corner is now 1, which divides everything, so no division step is needed.

*Clear the first column.* Add  $-(x-2)$  times row 1 to row 2, and add row 1 to row 3. In row 2 this gives entries  $-1 + (x-2)^2 = (x-1)(x-3)$  and  $2 + 2(x-2) = 2(x-1)$ ; in

row 3 it gives  $-1 - (x - 2) = -(x - 1)$  and  $x + 1 - 2 = x - 1$ :

$$\begin{pmatrix} 1 & -(x-2) & -2 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix}.$$

*Clear the first row.* Add  $(x - 2)$  times column 1 to column 2, and 2 times column 1 to column 3. Since the first column below the corner is zero, nothing else changes:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix} = \text{diag}(1) \oplus \begin{pmatrix} (x-1)(x-3) & 2(x-1) \\ -(x-1) & x-1 \end{pmatrix}.$$

Every entry of the  $2 \times 2$  block is divisible by 1, so we move on to the block.

*The block.* Its entries of least degree are the three of degree 1. Swap rows 2 and 3 and scale the new row 2 by  $-1$ , putting the monic  $x - 1$  in the corner of the block:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & -(x-1) \\ 0 & (x-1)(x-3) & 2(x-1) \end{pmatrix}.$$

Now  $x - 1$  divides every entry of its row and column. Add column 2 to column 3, which clears the block's first row and turns the  $(3, 3)$ -entry into  $2(x - 1) + (x - 1)(x - 3) = (x - 1)(x - 1)$ :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & 0 \\ 0 & (x-1)(x-3) & (x-1)^2 \end{pmatrix}.$$

Add  $-(x - 3)$  times row 2 to row 3 to clear the block's first column:

$$D = \text{diag}(1, x - 1, (x - 1)^2),$$

and  $1 \mid x - 1 \mid (x - 1)^2$ , so this is the Smith normal form.

*The check.* The matrix  $xI - A$  has an entry  $-1$ , so  $\Delta_1 = 1$ . Its determinant is  $p_A = (x - 1)^3$  — expanding along the first row,

$$(x-2)[(x-2)(x+1)+2] + [-(x+1)+2] + 2[1+(x-2)] = (x-2)(x^2-x) + (1-x) + 2(x-1),$$

which is  $(x-2)x(x-1) + (x-1) = (x-1)(x^2-2x+1) = (x-1)^3$  — so  $\Delta_3 = (x-1)^3$ . For  $\Delta_2$ , the minor on rows  $\{2, 3\}$  and columns  $\{1, 2\}$  is  $\det \begin{pmatrix} -1 & x-2 \\ -1 & -1 \end{pmatrix} = 1 + (x-2) = x-1$ , so  $\Delta_2 \mid x-1$ ; and every  $2 \times 2$  minor is divisible by  $x-1$ , as Claim 2 predicts from  $D$ . Hence  $\Delta_2 = x-1$ , and

$$(d_1, d_2, d_3) = \left( \Delta_1, \frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_2} \right) = (1, x - 1, (x - 1)^2),$$

in agreement. As a sanity check on  $A$  itself:  $A - I$  has rank 1, so  $g_A(1) = 2$  and the Jordan form of  $A$  is  $J_2(1) \oplus J_1(1)$  (Corollary 9.3.2).

### 9.7.4 Similarity read off from a polynomial matrix

We can now state the theorem that makes all of this useful. It converts a question about  $A$  and  $B$  — is there an invertible  $P \in M_n(F)$  with  $B = P^{-1}AP$ ? — into a question about the two polynomial matrices  $xI - A$  and  $xI - B$ , which the Smith algorithm answers by computation.

One direction is immediate. The other needs a tool: polynomial matrices can be divided by  $xI - A$ , with a **constant** remainder.

#### Lemma 9.7.4 — Division of a Polynomial Matrix by $xI - A$

Let  $A \in M_n(F)$  and  $M \in M_n(F[x])$ . Then there are  $Q \in M_n(F[x])$  and  $R \in M_n(F)$  with

$$M = (xI - A)Q + R.$$

*Idea.* Write  $M$  as a polynomial  $\sum_k M_k x^k$  with **constant matrix** coefficients. The scalar identity  $x^k - a^k = (x - a)(x^{k-1} + \dots + a^{k-1})$  has a matrix version  $x^k I - A^k = (xI - A)S_k$ , in which the second factor is a polynomial in  $x$  and  $A$  and so commutes with  $xI - A$ . Replacing each  $x^k I$  by  $(x^k I - A^k) + A^k$  puts a factor  $xI - A$  in front of everything except a constant matrix.

*Proof.* Collecting the coefficients of each power of  $x$  in the entries of  $M$ , write  $M = \sum_{k=0}^m x^k M_k$  with  $M_0, \dots, M_m \in M_n(F)$ ; note that  $x^k$  is a scalar polynomial, so  $x^k M_k = M_k x^k$ . For  $k \geq 1$  put

$$S_k := \sum_{j=0}^{k-1} x^{k-1-j} A^j \in M_n(F[x]), \quad S_0 := 0.$$

Expanding the product and telescoping,

$$(xI - A)S_k = \sum_{j=0}^{k-1} x^{k-j} A^j - \sum_{j=0}^{k-1} x^{k-1-j} A^{j+1} = x^k I - A^k$$

for every  $k \geq 0$ . Therefore

$$M = \sum_{k=0}^m (x^k I) M_k = \sum_{k=0}^m (x^k I - A^k) M_k + \sum_{k=0}^m A^k M_k = (xI - A) \underbrace{\sum_{k=0}^m S_k M_k}_{=Q} + \underbrace{\sum_{k=0}^m A^k M_k}_{=R},$$

with  $Q \in M_n(F[x])$  and  $R \in M_n(F)$ , as claimed. ■

#### Theorem 9.7.5 — Similarity as Equivalence of $xI - A$ and $xI - B$

Let  $F$  be a field,  $n \geq 1$  and  $A, B \in M_n(F)$ . Then

$$A \sim B \text{ over } F \quad \Longleftrightarrow \quad xI - A \approx xI - B \text{ over } F[x].$$

*Idea.*  $(\Rightarrow)$  is a one-line computation: a constant invertible  $P$  is also an invertible polynomial matrix.

$(\Leftarrow)$  is the substance, and the whole difficulty is that the matrices  $U, V$  witnessing the equivalence have polynomial entries, while we must produce a **constant**  $P$ . The trick is to divide  $U$  by  $xI - B$  and keep only the constant remainder  $P$ . Comparing degrees in the resulting identity shows that  $P$  alone already conjugates:  $P(xI - A) = (xI - B)P$ , that is,  $PA = BP$ . Running the same argument on the inverted relation produces a second constant  $P'$  with  $P'B = AP'$ , and multiplying the two expressions for  $V^{-1}$  and  $V$  shows  $PP' = I$ , so  $P$  is invertible.

*Proof.*  $(\Rightarrow)$  Suppose  $B = P^{-1}AP$  with  $P \in M_n(F)$  invertible. Then, inside  $M_n(F[x])$ ,

$$xI - B = P^{-1}(xI)P - P^{-1}AP = P^{-1}(xI - A)P,$$

and  $P^{-1}, P \in M_n(F) \subseteq M_n(F[x])$  have non-zero constant determinants, hence are invertible over  $F[x]$  (Proposition 9.7.1). So  $xI - A \approx xI - B$ .

$(\Leftarrow)$  Suppose

$$xI - B = U(xI - A)V \tag{*}$$

with  $U, V \in M_n(F[x])$  invertible over  $F[x]$ .

**Step 1. Divide  $U$  by  $xI - B$ .** By Lemma 9.7.4 there are  $Q \in M_n(F[x])$  and  $P \in M_n(F)$  with  $U = (xI - B)Q + P$ . Substituting into (\*),

$$xI - B = (xI - B)Q(xI - A)V + P(xI - A)V,$$

so  $P(xI - A)V = (xI - B)[I - Q(xI - A)V]$ . Multiplying on the right by  $V^{-1}$  and putting

$$S := V^{-1} - Q(xI - A) \in M_n(F[x]),$$

we get

$$P(xI - A) = (xI - B)S. \tag{1}$$

**Step 2.  $S$  is a constant matrix, and  $PA = BP$ .** Suppose  $S \neq 0$  and write  $S = \sum_{k=0}^d x^k S_k$  with  $S_k \in M_n(F)$  and  $S_d \neq 0$ . The coefficient of  $x^{d+1}$  in  $(xI - B)S$  is  $S_d \neq 0$ , while the left-hand side of (1) is  $xP - PA$ , whose coefficients vanish beyond  $x^1$ . Hence  $d+1 \leq 1$ , that is  $d = 0$ . So in every case  $S \in M_n(F)$ . Comparing the coefficients of  $x^1$  in (1) gives  $P = S$ , and comparing the constant terms gives  $-PA = -BP$ . Therefore

$$PA = BP, \quad \text{equivalently} \quad P(xI - A) = (xI - B)P, \tag{2}$$

the second form because  $P(xI - A) = xP - PA = xP - BP = (xI - B)P$ . Moreover  $S = P$  turns the definition of  $S$  into

$$V^{-1} = P + Q(xI - A). \tag{3}$$

**Step 3. The same argument, inverted.** From (\*),  $xI - A = U^{-1}(xI - B)V^{-1}$ , which is (\*) with  $A, B$  interchanged and  $U, V$  replaced by the invertible polynomial matrices  $U^{-1}, V^{-1}$ . So Steps 1 and 2 apply and give  $Q' \in M_n(F[x])$  and  $P' \in M_n(F)$  with

$$P'(xI - B) = (xI - A)P' \quad \text{and} \quad V = P' + Q'(xI - B). \tag{4}$$

**Step 4.  $P$  is invertible.** Multiplying (3) by (4),

$$I = V^{-1}V = [P+Q(xI-A)][P'+Q'(xI-B)] = PP' + PQ'(xI-B) + Q(xI-A)P' + Q(xI-A)Q'(xI-B).$$

By the first identity in (4),  $(xI-A)P' = P'(xI-B)$ , so the third summand is  $QP'(xI-B)$  and

$$I - PP' = Y(xI - B), \quad Y := PQ' + QP' + Q(xI - A)Q' \in M_n(F[x]).$$

If  $Y \neq 0$ , write  $Y = \sum_{k=0}^e x^k Y_k$  with  $Y_e \neq 0$ ; the coefficient of  $x^{e+1}$  on the right is  $Y_e \neq 0$ , while the left-hand side is constant. Hence  $Y = 0$  and  $PP' = I$ . By Theorem 2.4.5,  $P$  is invertible with  $P^{-1} = P'$ .

**Step 5. Conclude.** By (2),  $PA = BP$ , so  $B = PAP^{-1}$  and  $A \sim B$ . This proves the theorem. ■

Two by-products are worth recording. Whenever  $xI - B = U(xI - A)V$  with  $U, V$  invertible over  $F[x]$ , taking determinants gives  $p_B = (\det U)(\det V) p_A$  with  $\det U, \det V$  non-zero constants (Proposition 9.7.1); both  $p_A$  and  $p_B$  are monic of degree  $n$ , so

$$p_A = p_B \quad \text{and} \quad (\det U)(\det V) = 1,$$

which recovers the invariance of the characteristic polynomial. And, by Theorem 9.7.3, the test is now decidable by a finite computation in  $F[x]$ : reduce  $xI - A$  and  $xI - B$  to Smith normal form and compare.

### Corollary 9.7.6 — The Smith Form of $xI - A$ Carries the Invariant Factors

Let  $A \in M_n(F)$  with invariant factors  $d_1 \mid d_2 \mid \cdots \mid d_r$  as in Theorem 9.6.5. Then the Smith normal form of  $xI - A$  is

$$\text{diag} \left( \underbrace{1, \dots, 1}_{n-r}, d_1, d_2, \dots, d_r \right).$$

In particular the invariant factors of  $A$  are exactly the non-constant diagonal entries of the Smith normal form of  $xI - A$ , read from the top down, and two matrices in  $M_n(F)$  are similar if and only if  $xI - A$  and  $xI - B$  have the same Smith normal form.

*Proof.* By Theorem 9.6.5,  $A \sim C(d_1) \oplus \cdots \oplus C(d_r)$  with  $d_i$  monic,  $\deg d_i = k_i \geq 1$  and  $\sum k_i = n$ . By the ( $\Rightarrow$ ) direction of Theorem 9.7.5 it is enough to find the Smith normal form of

$$xI_n - (C(d_1) \oplus \cdots \oplus C(d_r)) = (xI_{k_1} - C(d_1)) \oplus \cdots \oplus (xI_{k_r} - C(d_r)),$$

the equality by Theorem 7.1.4 (a).

*One block.* Let  $d$  be monic of degree  $k \geq 1$ . By Definition 9.5.3, the only non-zero entries of  $xI_k - C(d)$  are  $x$  on the diagonal (with  $x + a_{k-1}$  in the corner),  $-1$  in each position  $(i+1, i)$ , and the entries of the last column. Delete row 1 and column  $k$ . In the remaining  $(k-1) \times (k-1)$  matrix, the entry in position  $(p, q)$  is the  $(p+1, q)$ -entry of  $xI_k - C(d)$ : it is  $-1$  when  $q = p$ , it is  $x$  when  $q = p+1$ , and it is 0 when  $q < p$ .

So that matrix is upper triangular with every diagonal entry  $-1$ , and its determinant is  $(-1)^{k-1} \neq 0$  by Theorem 6.4.6, again over  $F[x]$ . Hence  $\Delta_{k-1}(xI_k - C(d)) = 1$ . By Claim 2 in the proof of Theorem 9.7.3, the determinantal divisors of any matrix are the partial products  $d_1 \cdots d_j$  of its Smith diagonal, so  $\Delta_j \mid \Delta_{k-1} = 1$  and therefore  $\Delta_j = 1$  for every  $j \leq k-1$ . Also  $\Delta_k = \det(xI_k - C(d)) = p_{C(d)} = d$  by Theorem 9.5.3. By the uniqueness part of Theorem 9.7.3, the Smith normal form of  $xI_k - C(d)$  is  $\text{diag}(1, \dots, 1, d)$ . (For  $k = 1$  read this as  $\text{diag}(d)$ .)

*Assembling.* If  $M_i \approx M'_i$  with witnesses  $U_i, V_i$ , then  $\bigoplus_i M_i \approx \bigoplus_i M'_i$  with witnesses  $\bigoplus_i U_i$  and  $\bigoplus_i V_i$ , which are invertible with inverses  $\bigoplus_i U_i^{-1}, \bigoplus_i V_i^{-1}$  (Theorem 7.1.4 (a), (c)). So  $xI_n - A$  is equivalent to the direct sum of the matrices  $\text{diag}(1, \dots, 1, d_i)$ , a diagonal matrix whose diagonal entries are  $n - r$  ones and the polynomials  $d_1, \dots, d_r$ . Simultaneous row and column swaps — each a pair of elementary operations — reorder these entries at will, so  $xI_n - A$  is equivalent to

$$\text{diag}(1, \dots, 1, d_1, d_2, \dots, d_r),$$

and this is already in the shape required by Theorem 9.7.3 (a), because  $1 \mid d_1 \mid d_2 \mid \cdots \mid d_r$  and all the entries are monic. By uniqueness it *is* the Smith normal form of  $xI - A$ . The final claim combines this with Theorem 9.7.5 and the uniqueness in Theorem 9.7.3. ■

So the determinantal divisors of  $xI - A$  compute the invariant factors of  $A$  without any factorization of polynomials:  $\Delta_k$  is a gcd, obtained from the Euclidean algorithm alone. Moreover  $\Delta_n(xI - A) = p_A$  and  $\Delta_n/\Delta_{n-1} = d_r = m_A$ .

## 9.7.5 Every matrix is similar to its transpose

Chapter 3 asked whether  $A$  and  $A^\top$  are always similar and postponed the answer; Exercise (C1: A matrix is similar to its transpose) settled it whenever  $p_A$  splits. The Smith form removes that hypothesis, because transposing a matrix does not change its minors.

### Corollary 9.7.7 — A Matrix Is Similar to Its Transpose

Let  $F$  be any field and  $A \in M_n(F)$ . Then  $A \sim A^\top$ .

*Proof.* Let  $M = xI - A$ , so that  $M^\top = xI - A^\top$ . For  $k$ -sets  $I, J$  we have  $(M^\top)_{I,J} = (M_{J,I})^\top$  by Definition 0.9.10, so  $\det(M^\top)_{I,J} = \det M_{J,I}$  by Theorem 6.4.5, valid over the commutative ring  $F[x]$ . Hence the  $k \times k$  minors of  $M^\top$  are exactly the  $k \times k$  minors of  $M$ , listed in a different order, and

$$\Delta_k(xI - A^\top) = \Delta_k(xI - A) \quad (k = 1, \dots, n).$$

By Theorem 9.7.3, each of  $xI - A$  and  $xI - A^\top$  is equivalent over  $F[x]$  to the diagonal matrix built from its own determinantal divisors, and these two diagonal matrices coincide. Since  $\approx$  is an equivalence relation (Proposition 9.7.2 (a)),  $xI - A \approx xI - A^\top$ . By Theorem 9.7.5,  $A \sim A^\top$ . ■

Two sentences of consequence. An operator and its dual have the same invariants: by Theorem 4.4.2 the matrix of  $T'$  in the dual basis is  $[T]_B^T$ , so  $T$  and  $T'$  share their characteristic polynomial, their minimal polynomial and their invariant factors. And a matrix and its transpose always have the same Jordan form when one exists, which says that the *rows* of  $A - \lambda I$  are exactly as informative as its columns.

**Warning.** **Similar is not the same as “equivalent over  $F[x]$ ” for the matrices themselves.** The theorem compares  $xI - A$  with  $xI - B$ , not  $A$  with  $B$ . As polynomial matrices, any two invertible  $A, B \in M_n(F)$  are equivalent over  $F[x]$  — take  $U = A^{-1}$ ,  $V = B$  — whether or not they are similar. The whole content of Theorem 9.7.5 is that inserting the variable  $x$  on the diagonal remembers the conjugation.

## 9.7.6 Exercises

### 9.7.6.1 A. Check your understanding

#### Exercise A1.

- List the three elementary operations over  $F[x]$ , and say which one differs from its counterpart over a field and why.
- State a criterion for a matrix  $U \in M_n(F[x])$  to be invertible in  $M_n(F[x])$ .
- State the Smith Normal Form theorem, including the uniqueness clause.
- Define the  $k$ -th determinantal divisor of a polynomial matrix, and say why it is unchanged by equivalence.
- True or false: if  $\det(xI - A) = \det(xI - B)$ , then  $xI - A \approx xI - B$ . Justify your answer.
- How do you read  $m_A$  off the Smith normal form of  $xI - A$ ?

*Solution.*

- Swap two rows; multiply a row by a non-zero **scalar** of  $F$ ; add a **polynomial** multiple of one row to a different row (Definition 9.7.1). The scaling operation is restricted, because only the non-zero constants are invertible in  $F[x]$ ; multiplying a row by  $x$  is not reversible.
- $U$  is invertible over  $F[x]$  if and only if  $\det U$  is a non-zero constant (Proposition 9.7.1).
- Every  $M \in M_n(F[x])$  is equivalent over  $F[x]$ , by elementary operations, to a diagonal  $\text{diag}(d_1, \dots, d_n)$  with each  $d_i$  monic or zero and  $d_i \mid d_{i+1}$ ; this diagonal is unique (Theorem 9.7.3).
- $\Delta_k(M)$  is the monic generator of the ideal generated by all  $k \times k$  minors of  $M$ , equivalently their monic gcd. It is unchanged by equivalence because every  $k \times k$  minor of  $UMV$  is an  $F[x]$ -combination of  $k \times k$  minors of  $M$ , by the Cauchy–Binet formula (Claim 1 in the proof of Theorem 9.7.3).
- False. Equality of determinants says only  $\Delta_n(xI - A) = \Delta_n(xI - B)$ ; the lower determinantal divisors can differ. For  $A = J_2(0) \oplus J_2(0)$  and  $B = J_3(0) \oplus J_1(0)$  in  $M_4(F)$ , both determinants are  $x^4$ , but the Smith normal forms are  $\text{diag}(1, 1, x^2, x^2)$  and  $\text{diag}(1, 1, x, x^3)$ . (Equivalently, the matrices are not similar, by Corollary 9.3.3.)

- (f)  $m_A$  is the **last** diagonal entry of the Smith normal form of  $xI - A$ , that is  $\Delta_n/\Delta_{n-1} = p_A/\Delta_{n-1}$  (Corollary 9.7.6, since the last diagonal entry is  $d_r = m_A$  in the notation there).

### 9.7.6.2 B. Practice

#### Exercise B1: Smith normal forms.

Find the Smith normal form over  $\mathbb{Q}[x]$  of each of the following, naming the operations you use.

- (a)  $M_1 = \begin{pmatrix} x & x+1 \\ x-1 & x \end{pmatrix}.$
- (b)  $M_2 = \begin{pmatrix} x^2-1 & 0 \\ 0 & x^2-x \end{pmatrix}.$
- (c)  $M_3 = \begin{pmatrix} x & 0 & 1 \\ -1 & x & 0 \\ 0 & -1 & x \end{pmatrix}.$

*Solution.*

- (a)  $\det M_1 = x^2 - (x+1)(x-1) = 1$ , so  $M_1$  is invertible over  $\mathbb{Q}[x]$  (Proposition 9.7.1) and  $\Delta_1 = \Delta_2 = 1$ . The Smith normal form is  $\text{diag}(1, 1) = I_2$ . (Concretely: subtract column 1 from column 2 to get  $\begin{pmatrix} x & 1 \\ x-1 & 1 \end{pmatrix}$ , then subtract row 1 from row 2, swap columns, and clear.)
- (b) Here  $\Delta_1 = \gcd(x^2-1, x^2-x) = \gcd((x-1)(x+1), x(x-1)) = x-1$  and  $\Delta_2 = \det M_2 = x(x-1)^2(x+1)$ . So the Smith normal form is

$$\text{diag}\left(x-1, \frac{x(x-1)^2(x+1)}{x-1}\right) = \text{diag}(x-1, x(x-1)(x+1)),$$

and indeed  $x-1$  divides  $x(x-1)(x+1)$ . By operations: add column 2 to column 1, giving first column  $(x^2-1, x^2-x)$ ; subtract row 1 from row 2 to get  $(x^2-1, 1-x)$  in column 1; the entry  $1-x$  has least degree, so swap it into the corner, scale by  $-1$ , and continue.

- (c) The  $2 \times 2$  minor on rows  $\{2, 3\}$  and columns  $\{1, 2\}$  is  $\det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} = 1$ , so  $\Delta_2 = 1$ , and then  $\Delta_1 = 1$  as well (it divides  $\Delta_2$ ). Expanding  $\det M_3$  along the first row,  $x(x^2-0) + 1 \cdot (1-0) = x^3+1$ , so  $\Delta_3 = x^3+1$  and the Smith normal form is  $\text{diag}(1, 1, x^3+1)$ . Note  $M_3 = xI - C$  with  $C = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = C(x^3+1)$ , and Corollary 9.7.6 predicts exactly this, since a companion matrix has the single invariant factor  $x^3+1$ .

#### Exercise B2: Invariant factors from determinantal divisors.

For each matrix  $A$ , compute  $\Delta_1, \dots, \Delta_n$  of  $xI - A$ , write down the Smith normal form, and read off the invariant factors,  $m_A$  and the Jordan form.

- (a)  $A = J_2(0) \oplus J_2(0) \in M_4(\mathbb{Q})$ .  
 (b)  $A = J_3(0) \oplus J_1(0) \in M_4(\mathbb{Q})$ .  
 (c)  $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{Q})$ .

*Solution.* One remark makes the minors easy. If  $M = M_1 \oplus M_2$  with square blocks, and  $I, J$  are  $k$ -sets, split  $I = I_1 \cup I_2$  and  $J = J_1 \cup J_2$  according to the blocks. If  $|I_1| \neq |J_1|$ , then  $M_{I,J}$  has a zero row or a zero column and  $\det M_{I,J} = 0$ ; if  $|I_1| = |J_1|$ , then  $M_{I,J} = (M_1)_{I_1,J_1} \oplus (M_2)_{I_2,J_2}$  and the determinant is the product of the two, by Theorem 6.8.4, whose proof uses only sums and products of entries and so applies over  $F[x]$  (the remark after that theorem). So **the minors of a block diagonal matrix are the products of matching-size minors of the blocks**. Here  $xI - A$  is the direct sum of the blocks  $xI_k - J_k(0)$ , each upper triangular with  $x$  on the diagonal and  $-1$  on the superdiagonal.

- (a)  $xI - A = N \oplus N$  with  $N = \begin{pmatrix} x & -1 \\ 0 & x \end{pmatrix}$ , whose minors are the entries  $x, -1, 0$  for  $k = 1$  and  $x^2$  for  $k = 2$ . By the remark, the minors of  $N \oplus N$  are  $\{x, -1, 0\}$  for  $k = 1$ ; the products of two  $1 \times 1$  minors together with  $x^2$  for  $k = 2$ ;  $x^2$  times a  $1 \times 1$  minor for  $k = 3$ ; and  $x^4$  for  $k = 4$ . Hence

$$\Delta_1 = 1, \quad \Delta_2 = 1 \quad (\text{from } (-1)(-1) = 1), \quad \Delta_3 = x^2, \quad \Delta_4 = x^4,$$

so the Smith normal form is  $\text{diag}(1, 1, x^2, x^2)$ , the invariant factors are  $x^2 \mid x^2$ ,  $m_A = x^2$  is the last of them, and the Jordan form is  $J_2(0) \oplus J_2(0)$ .

- (b) Now  $xI - A = N_3 \oplus (x)$  with  $N_3 = \begin{pmatrix} x & -1 & 0 \\ 0 & x & -1 \\ 0 & 0 & x \end{pmatrix}$ . The minors of  $N_3$  are  $\{x, -1, 0\}$  for  $k = 1$ ;  $\{x^2, -x, 1, 0\}$  for  $k = 2$ , the value 1 coming from rows  $\{1, 2\}$  and columns  $\{2, 3\}$ ; and  $x^3$  for  $k = 3$ . By the remark, the  $k \times k$  minors of  $xI - A$  are those of  $N_3$  together with  $x$  times its  $(k-1) \times (k-1)$  minors, so

$$\Delta_1 = 1, \quad \Delta_2 = \gcd(x^2, -x, 1, x \cdot x, -x) = 1, \quad \Delta_3 = \gcd(x^3, x \cdot x^2, -x \cdot x, x \cdot 1) = x, \quad \Delta_4 = x^4.$$

The Smith normal form is  $\text{diag}(1, 1, x, x^3)$ : the invariant factors are  $x \mid x^3$ ,  $m_A = x^3$  is the last of them, and the Jordan form is  $J_3(0) \oplus J_1(0)$ . Compared with (a) the determinants agree but  $\Delta_3$  does not, which is exactly why the two matrices are not similar.

- (c)  $xI - A = \begin{pmatrix} x-3 & -1 \\ 0 & x-3 \end{pmatrix}$ . Then  $\Delta_1 = 1$  and  $\Delta_2 = (x-3)^2$ , so the Smith normal form is  $\text{diag}(1, (x-3)^2)$ , the only invariant factor is  $(x-3)^2 = m_A = p_A$ , and the Jordan form is  $J_2(3)$ . Consistently,  $A$  is a companion matrix up to similarity, since  $m_A = p_A$  (Theorem 9.5.6).

### Exercise B3: Deciding similarity.

Determine whether each pair is similar over the stated field. Justify your answer using Theorem 9.7.5.

- (a)  $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .
- (b)  $A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}$  and  $B = J_2(1) \oplus J_1(1)$  over  $\mathbb{Q}$ .
- (c)  $A = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$  and  $B = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .

*Solution.*

- (a) Similar. Here  $B = A^\top$ , so [Corollary 9.7.7](#) applies. (Explicitly,  $\Delta_1 = 1$  and  $\Delta_2 = x^2 + 1$  for both, so both Smith normal forms are  $\text{diag}(1, x^2 + 1)$ .)
- (b) Similar. By [Example \(A Smith Normal Form Computed by Hand\)](#) the Smith normal form of  $xI - A$  is  $\text{diag}(1, x - 1, (x - 1)^2)$ . For  $B$ ,  $xI - B = \begin{pmatrix} x - 1 & -1 \\ 0 & x - 1 \end{pmatrix} \oplus (x - 1)$ . Its entry  $-1$  gives  $\Delta_1 = 1$ . By the block-minor rule in the solution to [Exercise \(B2: Invariant factors from determinantal divisors\)](#), the  $2 \times 2$  minors are  $(x - 1)^2$ , from the  $2 \times 2$  block, together with the products of a  $1 \times 1$  minor of each block, that is  $(x - 1)^2$ ,  $-(x - 1)$  and  $0$ ; so  $\Delta_2 = x - 1$ . Finally  $\Delta_3 = \det(xI - B) = (x - 1)^3$ . The Smith normal forms agree, so  $xI - A \approx xI - B$  and  $A \sim B$ .
- (c) Similar. Here

$$xI - A = \begin{pmatrix} x & 2 \\ -1 & x \end{pmatrix}, \quad xI - B = \begin{pmatrix} x & 1 \\ -2 & x \end{pmatrix},$$

and each has an entry that is a non-zero constant, so  $\Delta_1 = 1$  in both cases; and  $\Delta_2 = \det(xI - A) = x^2 + 2 = \det(xI - B)$ . So both Smith normal forms are  $\text{diag}(1, x^2 + 2)$ , hence  $xI - A \approx xI - B$  and  $A \sim B$  by [Theorem 9.7.5](#). Note that  $x^2 + 2$  is irreducible over  $\mathbb{Q}$ , so no Jordan form exists over  $\mathbb{Q}$  and [Exercise \(C1: A matrix is similar to its transpose\)](#) would not have decided this pair. (Here  $B \neq A^\top$ , so [Corollary 9.7.7](#) does not apply directly; the Smith forms settle it anyway.)

### 9.7.6.3 C. Going deeper

#### Exercise C1: The same algorithm over the integers.

The proof of [Theorem 9.7.3](#) used only one property of  $F[x]$ : division with remainder, where the “size” of a non-zero element is its degree. The integers  $\mathbb{Z}$  have the same property, with  $|a|$  in place of  $\deg$ .

- (a) State the version of [Theorem 9.7.3](#) for  $M \in M_n(\mathbb{Z})$  that the same proof gives, saying what replaces “monic”.
- (b) Compute the Smith normal form over  $\mathbb{Z}$  of  $M = \begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix}$ , naming the operations.
- (c) Every finite abelian group  $G$  with  $n$  generators is  $\mathbb{Z}^n$  modulo the column span of some  $M \in M_n(\mathbb{Z})$ , the matrix of relations, exactly as  $V$  was  $F[x]^n$  modulo the columns of  $xI - A$ . Using (b), identify the group with relation matrix  $M$  as a product of cyclic groups.

*Hint: for (c), a diagonal relation matrix  $\text{diag}(d_1, \dots, d_n)$  presents the product of the*

cyclic groups  $\mathbb{Z}/d_i\mathbb{Z}$ .

*Solution.*

- (a) Every  $M \in M_n(\mathbb{Z})$  can be brought by integer elementary operations — swap two rows or columns, multiply a row or column by  $\pm 1$  (the units of  $\mathbb{Z}$ ), add an **integer** multiple of one row or column to another — to a diagonal matrix  $\text{diag}(d_1, \dots, d_n)$  with every  $d_i \geq 0$  and  $d_i \mid d_{i+1}$ , and this diagonal is unique. “Monic” is replaced by “non-negative”, which is the normalization available because the units of  $\mathbb{Z}$  are  $\pm 1$ . Uniqueness again follows from the determinantal divisors  $\Delta_k$ , now non-negative integer gcds of the  $k \times k$  minors, using the Cauchy–Binet formula over the commutative ring  $\mathbb{Z}$ .
- (b) The least absolute value among the entries is 2, already in the corner. It divides 4 and 6. Subtract 2 times column 1 from column 2, and 3 times row 1 from row 2:

$$\begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 6 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix},$$

the last step multiplying row 2 by  $-1$ . Since  $2 \mid 8$ , the Smith normal form is  $\text{diag}(2, 8)$ . Check:  $\Delta_1 = \gcd(2, 4, 6, 4) = 2$  and  $\Delta_2 = |\det M| = |8 - 24| = 16 = 2 \cdot 8$ .

- (c) By (b) and the hint, the group is  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ , of order  $16 = |\det M|$ . This is the structure theorem for finite abelian groups in miniature: the invariant divisors  $d_1 \mid \dots \mid d_n$  of the relation matrix are the orders of the cyclic factors.

### Exercise C2: Uniqueness by hand in the smallest case.

Let  $M \in M_2(F[x])$  be non-zero.

- (a) Prove directly, without using determinantal divisors, that  $\Delta_1(M)$  — the monic gcd of the four entries of  $M$  — is unchanged by each of the six elementary operations of [Definition 9.7.1](#).
- (b) Prove that  $\det(UMV) = c \det M$  for a non-zero constant  $c$  whenever  $U, V$  are invertible over  $F[x]$ .
- (c) Deduce that if  $\text{diag}(d_1, d_2) \approx \text{diag}(e_1, e_2)$  with all four entries monic or zero,  $d_1 \mid d_2$  and  $e_1 \mid e_2$ , then  $d_1 = e_1$  and  $d_2 = e_2$ .

*Solution.*

- (a) Write  $d = \Delta_1(M)$ , the monic gcd of the entries, that is, the monic generator of the ideal  $\langle m_{11}, m_{12}, m_{21}, m_{22} \rangle$  (Theorem 5.2.2). A swap permutes the entries, so the ideal is unchanged. Scaling a row by  $c \in F \setminus \{0\}$  replaces two generators  $g$  by  $cg$ , and  $\langle cg \rangle = \langle g \rangle$  since  $c$  is a unit, so the ideal is unchanged. Adding  $h$  times row 1 to row 2 replaces  $m_{2j}$  by  $m_{2j} + hm_{1j}$ , which lies in the old ideal; so the new ideal is contained in the old one, and the operation is reversed by adding  $-h$  times row 1 to row 2, which gives the reverse containment. The same three arguments apply to columns. Hence  $d$  is unchanged.
- (b) By multiplicativity of the determinant over the commutative ring  $F[x]$  (Theorem 6.5.4 and the remark on commutative rings),  $\det(UMV) = \det U \cdot \det M \cdot \det V$ , and  $\det U, \det V$  are non-zero constants by [Proposition 9.7.1](#). Take  $c = \det U \det V$ .

- (c) By [Proposition 9.7.2](#) and (a),  $\Delta_1$  is an invariant of equivalence as far as elementary operations go; and for a general equivalence  $M' = U M V$  the argument of (a) applies verbatim with the generators of the ideal replaced using  $m'_{ij} = \sum_{k,l} u_{ik} m_{kl} v_{lj} \in \langle m_{kl} \rangle$ , and symmetrically. For the two diagonal matrices,  $\Delta_1(\text{diag}(d_1, d_2)) = \gcd(d_1, d_2) = d_1$  since  $d_1 \mid d_2$  and  $d_1$  is monic (or  $d_1 = d_2 = 0$ ); likewise  $\Delta_1(\text{diag}(e_1, e_2)) = e_1$ . Hence  $d_1 = e_1$ . By (b),  $d_1 d_2 = c e_1 e_2$  for some non-zero constant  $c$ ; if  $d_1 = e_1 \neq 0$  then  $d_2 = c e_2$ , and two monic polynomials (or two zeros) differing by a constant factor are equal, so  $d_2 = e_2$ . If  $d_1 = e_1 = 0$ , then  $d_2 = e_2 = 0$  by the divisibility chain.